B1xxSeries chips

FlashUpgrade andKEYencryption

Instructions V1.1

--2019.08.23

content:

1. Introduction to the tool

2. Operating instructions for various working modes

3. List of error codes and problem handling

Four, update records

1. Introduction to the tool

Mass production tool set, mainly includingCodeData package processing tools, code encryption tools,FlashThe code upgrade tool, and the user data and password burning tool, etc., their names and functions are introduced as follows:

(1) MVAssistant(PCsoftware)

-------- **Integrated data package, conversion, import, export, download and other functions**

(3) USB_Encrypt_Dongle(Hardware-dongle)

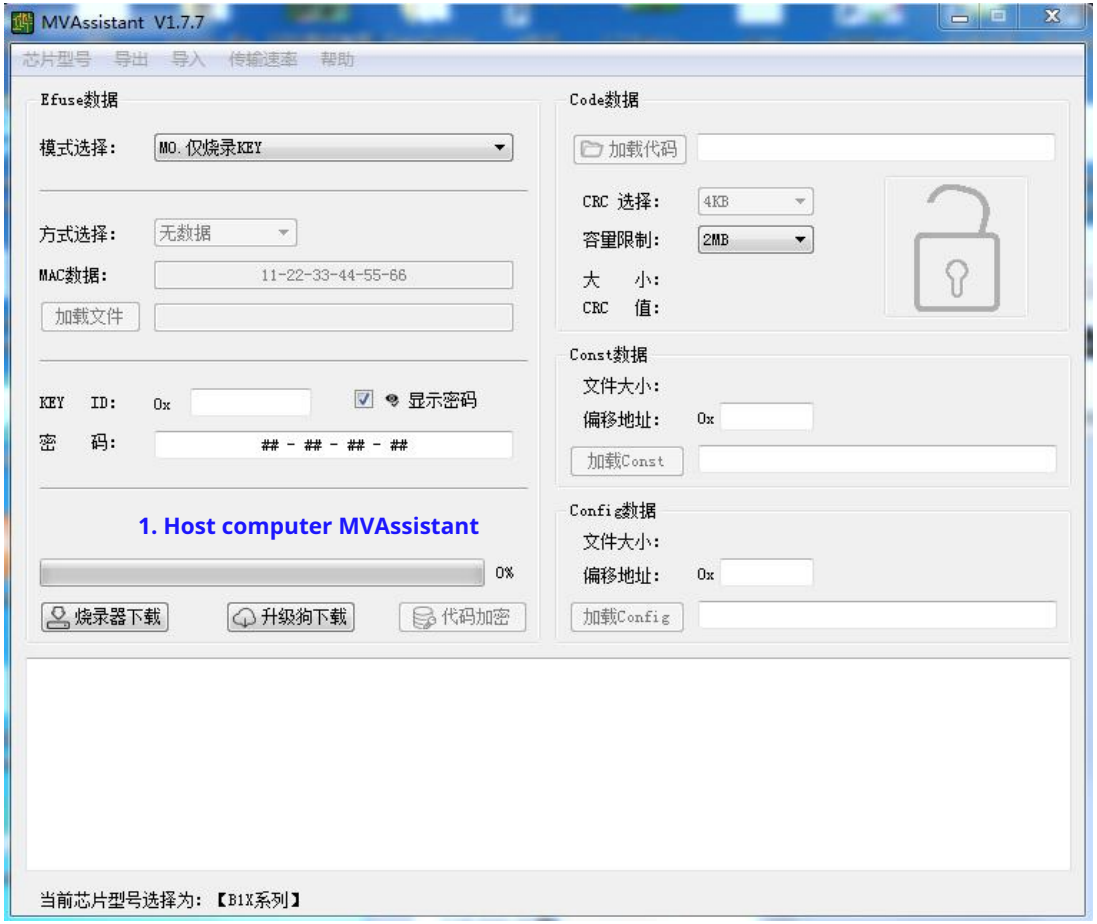
-------- **From the host computerCodeThe data is encrypted and returned to the host computer**

(4) Flash_Burner_Lite(Hardware-Upgrade Dog)

-------- **Receiving the data from the host computer, it can control the chip'sFlashTo upgrade**

(5) Efuse_Burner_Board(Hardware-Burner)

-------- **Receive from the host computerKEYAnd user data, the chip’sEfuseBurn**


MVAssistant V1.7.7
芯片型号 导出 导入 传输速率 帮助
Efuse数据
模式选择: M0.仅烧录KEY
方式选择: 无数据
MAC数据: 11-22-33-44-55-66
加载文件
KEY ID: 0x
显示密码
密 码: ## - ## - ## - ##
Code数据
加载代码
CRC 选择: 4KB
容里限制: 2MB
大 小:
CRC 值:
Const数据
文件大小:
偏移地址: 0x
加载Const
Config数据
文件大小:
偏移地址: 0x
加载Config
1. Host computer MVAssistant
0%
烧录器下载
升级狗下载
代码加密
当前芯片型号选择为: 【B1X系列】


3.加密狗 USB_Dongle_Encrypt


4.Flash升级工具 Flash_Burner_Lite


名称 修改日期 类型 大小
ALI 2018-5-6 16:31 文件夹
source 2018-5-25 16:45 文件夹
temp 2018-5-25 16:45 文件夹
all.bin BIN 文件 4 KB
MergeAudio2BinNew.exe 2018-4-26 11:29 应用程序 5,227 KB
o26_download_v0.9_IFLYTEK.exe 2018-4-17 14:19 应用程序 1,151 KB
sound_remind_item.h 2018-5-23 11:37 H 文件 1 KB
2.Const组包工具Merge_Bin_Tool


MVSILICON
START
5.烧录板 Efuse_Burner_Board

2. Operating instructions for various working modes

- Ø M0.Burn onlyKEY
- Ø M1.Burn onlyMAC Data M2.
- Ø Upgrade onlyFlash
- Ø M3.BurnKEY + BurnMAC Data M4.upgrade
- Ø Flash + BurnKEY M5.upgradeFlash + BurnMAC
- Ø Data M6.upgradeFlash + BurnKEY + BurnMAC
- Ø Data M7.【KEYcheck】
- Ø
- Ø M8.upgradeFlash + Equipment name
- Ø M9.upgradeFlash + BurnKEY +Equipment name

ShareM0~M9Ten working modes, which can complete operations such as chip programming, upgrading, and Bluetooth device information parameter writing:

1.BurnKEYRefers to the chipEfuseBurn user-defined4 Byte KEYvalue;

2.BurnMAC DataRefers to the chipFlashBurn the Bluetooth address and name in the designated location;

3.upgradeflashRefers to the development of customersCodeData upgrade to chipflashmiddle;

4.BurnMAC DataYesBT MAC,BLE MACAddress burned intoflashmiddle;

5.The name of the burning device isBT MAC,BLE MACBurn the device name intoflashmiddle;

Ten modesCan be selected according to the actual needs of customers.

The operation steps are summarized as follows:

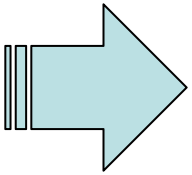
(Here is how to useM6Mode burn and upgrade, and load Bluetooth address and name)

(1) Connect the burner to the computer and supply power to the burner to ensure that it is connected to the computerUSBconnection succeeded(2) Select chipB1XSeries, and then tick in the mode selection columnM6 model(3) A dialog box pops up to set the Bluetooth name, fill in and click confirm to loadMACdata(4) Open the host computer tool to make various settings, and then download the mode and data to the burner (5) Load the customer code, insert the dongle, and encrypt the code (encryption is optional) (6)connectFlashUpgrade dog tool, downloadKeyorCodeTo the upgrade dog tool (7) Use the burner andFlashThe upgrade dog tool burns and upgrades customer chips

The following is a detailed description of the most common usersM6Mode, the operation of other modes is similar, only a brief introduction.

M6.upgradeFlash+BurnKEY+BurnMAC Data

Instructions



M6.upgradeFlash+BurnKEY+BurnMAC DataSteps

Step1: Prepare the writer and chip, connect the writer to the computer

(1) Connect the power cord 9V~12V

(2)connectBtypeUSBWire to computer (3) Toggle the switch to power on the burner, and the digital tube displays-PC-, Which means that the burner and the computer are successfully connected





M6.upgradeFlash+BurnKEY+BurnMAC DataSteps

Step2: Open the host computer tool, set various contents, download mode and data

(1) Select the chip model asB1xSeries, there will be a prompt message in the status bar (2) Mode selection, if it is an upgradeFlash+BurnKey+BurnMAC Data, Please select a modeM6

(3) Bluetooth address, enter the Bluetooth address, subBTandBLE, If the name is the same, check the setting is the same, click OK

(4)MACData setting, method selection: file input after no data or fixed data, if the Bluetooth address is fixed data, thenMACFill in the data box with data; if variable data needs to load the file, click Load document

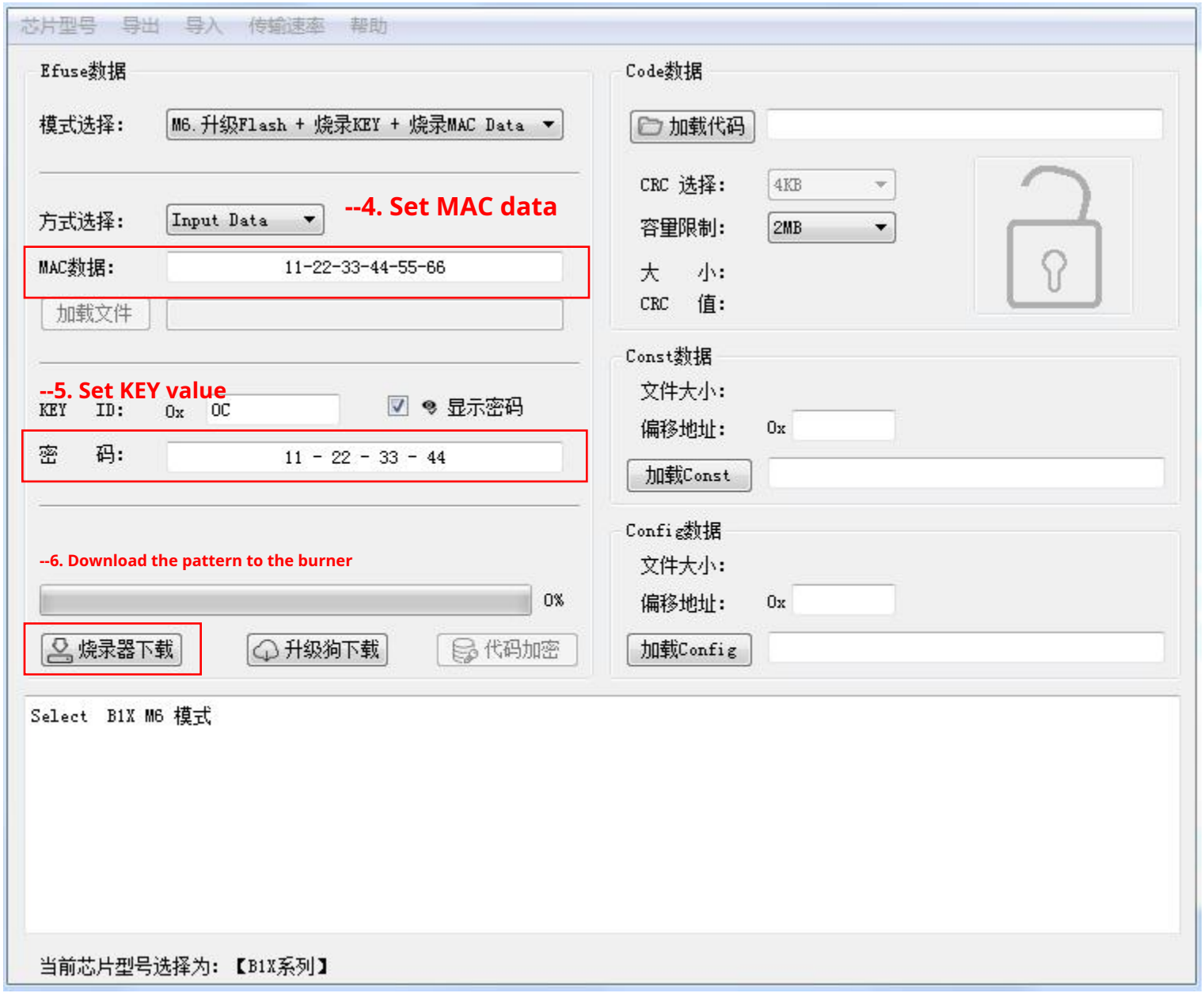
(5) Enter the password set by the user in the format4Hexadecimal data

(6) Click the [Burn Download] button to start downloading data to the writer, the download is complete,

Burner digital tube displayEndIndicates that the download was successful, and then unplugUSBLine, shownn-6 [instruction,nn-6Representation mode6, Because the digital tube can't displaym, Then usennTo representm]

As shown below:


MVAssistant V1.7.7
芯片型号
--1 Chip model selection
Efuse数据
模式选择:
M6.升级Flash + 烧录KEY + 烧录MAC Data
--2 Operation mode selection
方式选择:
无数据
Code数据
加载代码
CRC 选择:
4KB
容里限制:
2MB
请输入蓝牙设备名称
* 【BT 设备名称】 和 【BLE 设备名称】 最大长度 均为 40 Bytes
【BT 设备名称】:
--3Fill in the Bluetooth name
【BLE 设备名称】:
设置【BLE设备名称】与 【BT设备名称】 一样
--Set BT to be the same as BLE device name
确认
取消
当前芯片型号选择为: 【B1X系列】


芯片型号 导出 导入 传输速率 帮助
Efuse数据
模式选择: M6.升级Flash + 烧录KEY + 烧录MAC Data
方式选择: Input Data
--4. Set MAC data
MAC数据: 11-22-33-44-55-66
加载文件
--5. Set KEY value
KEY ID: 0x 0C
显示密码
密 码: 11 - 22 - 33 - 44
--6. Download the pattern to the burner
0%
烧录器下载
升级狗下载
代码加密
Code数据
加载代码
CRC 选择: 4KB
容里限制: 2MB
大 小:
CRC 值:
Const数据
文件大小:
偏移地址: 0x
加载Const
Config数据
文件大小:
偏移地址: 0x
加载Config
Select B1X M6 模式
当前芯片型号选择为:【B1X系列】

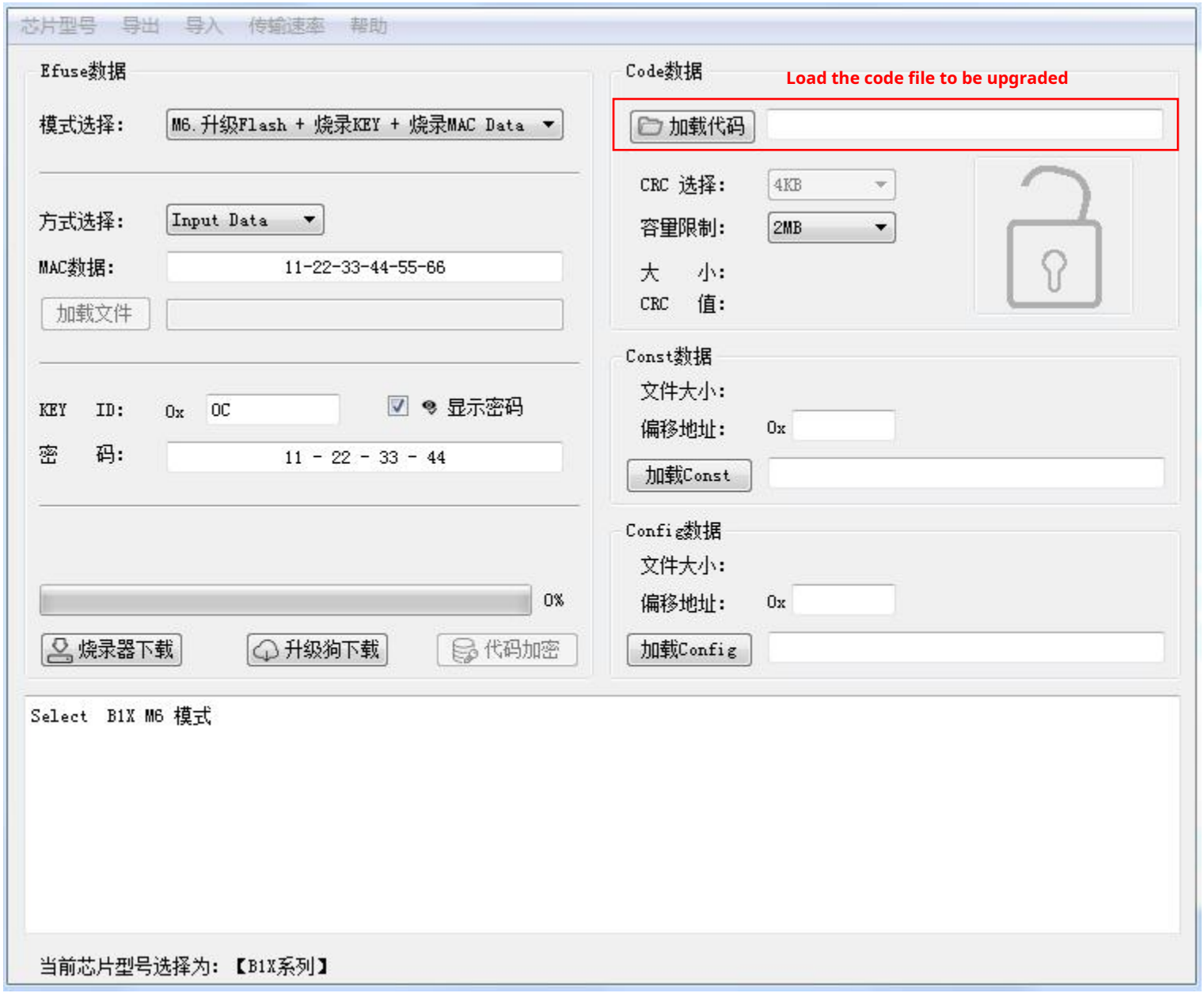
M6.upgradeFlash+BurnKEY+BurnMAC DataSteps

Step3: Ready to be upgradedCode

CodeDocument preparation

CodeThe file is created by the customer and generated after compiling the project.bindocument

The corresponding file loading method is as follows:


芯片型号
导出
导入
传输速率
帮助
Efuse数据
模式选择:
M6.升级Flash + 烧录KEY + 烧录MAC Data
方式选择:
Input Data
MAC数据:
11-22-33-44-55-66
加载文件
KEY ID:
0x
0C
显示密码
密 码:
11 - 22 - 33 - 44
0%
烧录器下载
升级狗下载
代码加密
Code数据
Load the code file to be upgraded
加载代码
CRC 选择:
4KB
容里限制:
2MB
大 小:
CRC 值:
Const数据
文件大小:
偏移地址:
0x
加载Const
Config数据
文件大小:
偏移地址:
0x
加载Config
Select B1X M6 模式
当前芯片型号选择为:【B1X系列】

M6.upgradeFlash+BurnKEY+BurnMAC DataSteps

Step4: For those to be upgradedFlashBinFile encryption

(Note that the chip model isB1Xseries)

(1)loadbinFile (demonstrated hereCodeFiles andConstWhen files are loaded at the same time)

[Note: File encryption is only forCodeContent is encrypted]

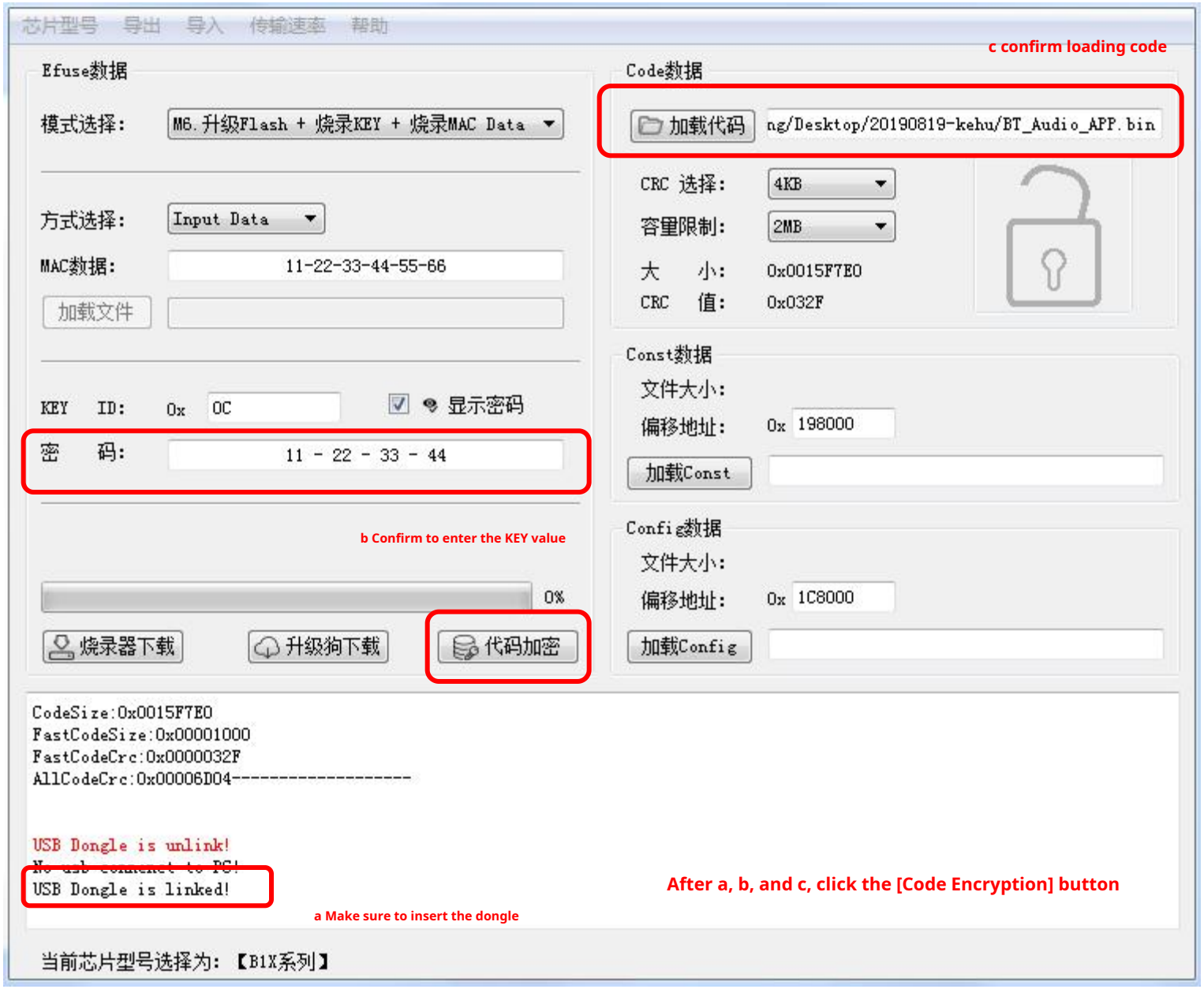
(2) Enter the password set by the user (entered before)

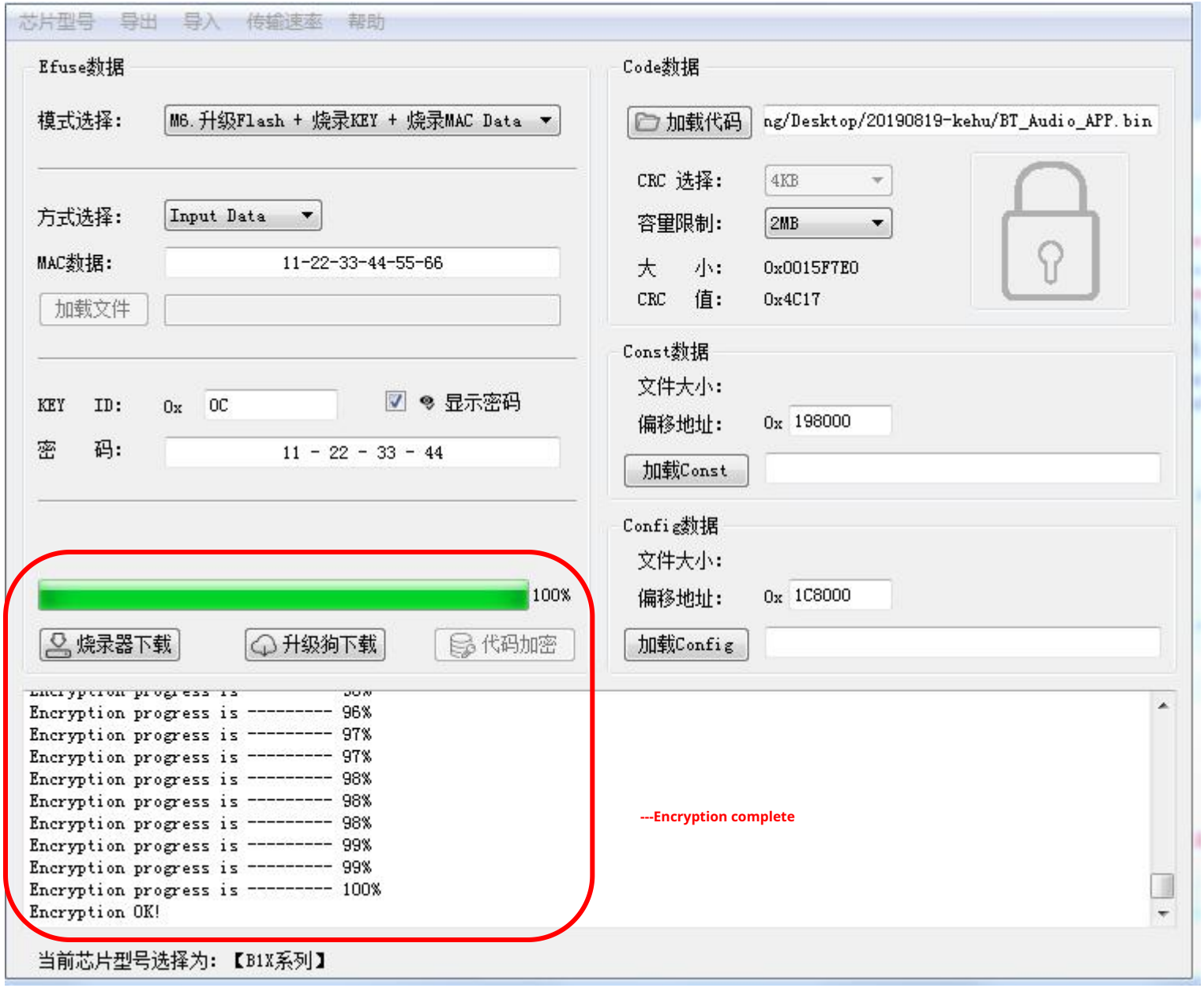
(3) Plug the dongle into the computer'sUSBinterface

(4) Click the button to start encryption and wait for the end of encryption



As shown below:


芯片型号 导出 导入 传输速率 帮助
c confirm loading code
Efuse数据
模式选择: M6.升级Flash + 烧录KEY + 烧录MAC Data
方式选择: Input Data
MAC数据: 11-22-33-44-55-66
加载文件
KEY ID: 0x 0C
显示密码
密 码: 11 - 22 - 33 - 44
b Confirm to enter the KEY value
0%
烧录器下载
升级狗下载
代码加密
Code数据
加载代码
ng/Desktop/20190819-kehu/BT_Audio_APP.bin
CRC 选择: 4KB
容里限制: 2MB
大 小: 0x0015F7E0
CRC 值: 0x032F
Const数据
文件大小:
偏移地址: 0x 198000
加载Const
Config数据
文件大小:
偏移地址: 0x 1C8000
加载Config
CodeSize:0x0015F7E0
FastCodeSize:0x00001000
FastCodeCrc:0x0000032F
AllCodeCrc:0x00006D04----------
USB Dongle is unlink!
USB Dongle is linked!
a Make sure to insert the dongle
After a, b, and c, click the [Code Encryption] button
当前芯片型号选择为: 【B1X系列】


芯片型号 导出 导入 传输速率 帮助
Efuse数据
模式选择: M6.升级Flash + 烧录KEY + 烧录MAC Data
方式选择: Input Data
MAC数据: 11-22-33-44-55-66
加载文件
KEY ID: 0x 0C
显示密码
密 码: 11 - 22 - 33 - 44
100%
烧录器下载
升级狗下载
代码加密
Code数据
加载代码
ng/Desktop/20190819-kehu/BT_Audio_APP.bin
CRC 选择: 4KB
容量限制: 2MB
大 小: 0x0015F7E0
CRC 值: 0x4C17
Const数据
文件大小:
偏移地址: 0x 198000
加载Const
Config数据
文件大小:
偏移地址: 0x 1C8000
加载Config
Encryption progress is --------- 96%
Encryption progress is --------- 97%
Encryption progress is --------- 97%
Encryption progress is --------- 98%
Encryption progress is --------- 98%
Encryption progress is --------- 98%
Encryption progress is --------- 99%
Encryption progress is --------- 99%
Encryption progress is --------- 100%
Encryption OK!
---Encryption complete
当前芯片型号选择为: 【B1X系列】

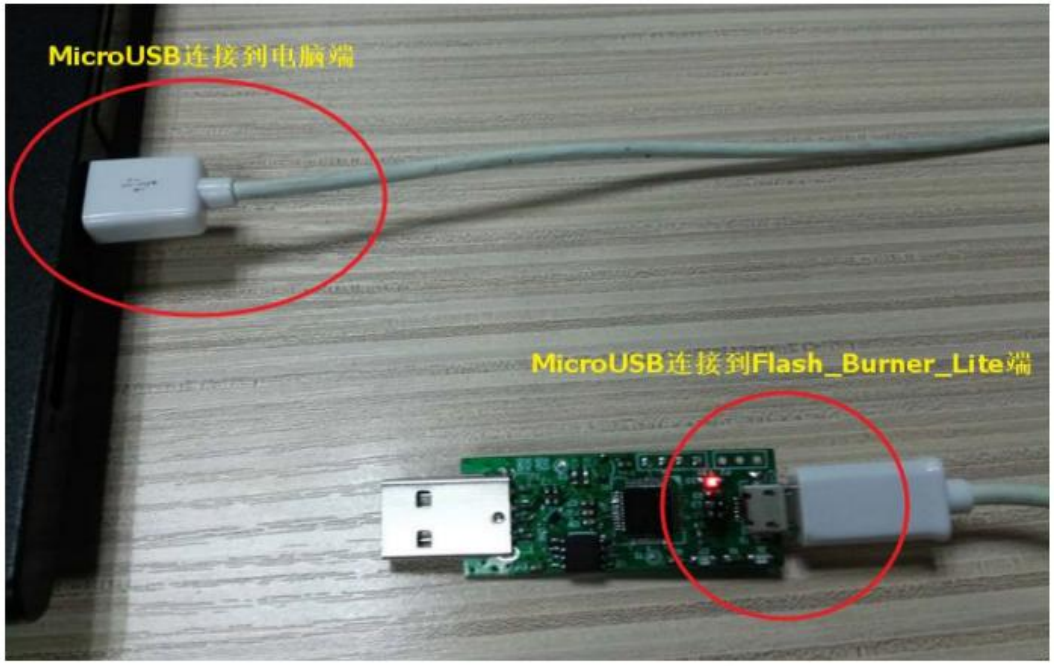
M6.upgradeFlash+BurnKEY+BurnMAC DataSteps

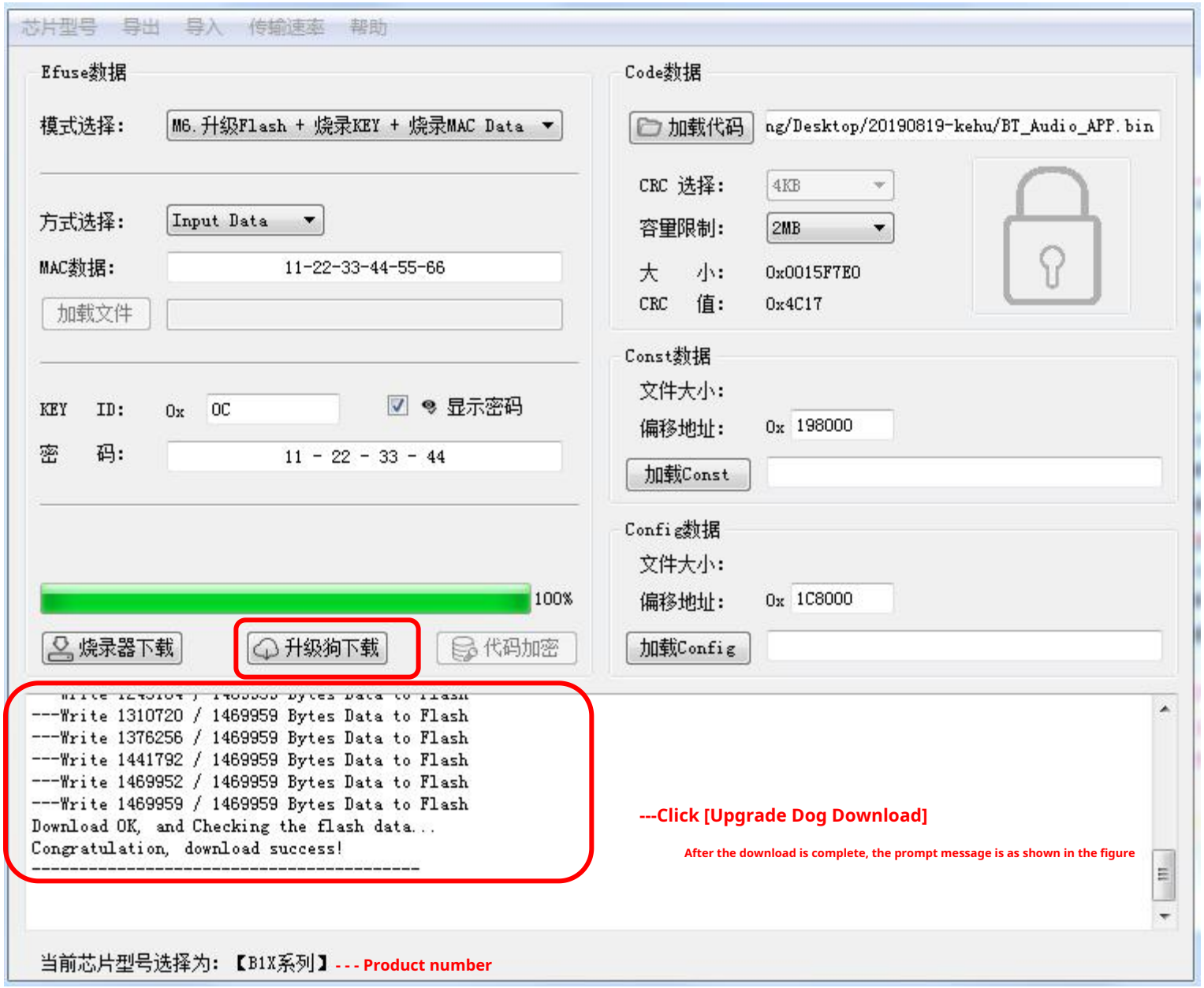
Step5:willKeyAnd encryptedCodeDownload toFlashUpgrade tool

(Upgrade dog tool: Flash_Burner_Lite **)**

(1) Hardware connection, useMicroUSBThe line connects the upgrade tool and the computer (2) Wait for the red and green status indicators of the upgrade tool to flash at the same time (initializationOK), the version number (3) Click on the host computer tool MVAssistantClick the [Upgrade Download] button to download,

Downloading, the green light flashes; the download is successful, the green light is always on, and the download fails, and the red light is always on.




芯片型号 导出 导入 传输速率 帮助
Efuse数据
模式选择: M6.升级Flash + 烧录KEY + 烧录MAC Data
方式选择: Input Data
MAC数据: 11-22-33-44-55-66
加载文件
KEY ID: 0x 0C
显示密码
密 码: 11 - 22 - 33 - 44
100%
烧录器下载
升级狗下载
代码加密
Code数据
加载代码
ng/Desktop/20190819-kehu/BT_Audio_APP.bin
CRC 选择: 4KB
容里限制: 2MB
大 小: 0x0015F7E0
CRC 值: 0x4C17
Const数据
文件大小:
偏移地址: 0x 198000
加载Const
Config数据
文件大小:
偏移地址: 0x 1C8000
加载Config
---Write 1310720 / 1469959 Bytes Data to Flash
---Write 1376256 / 1469959 Bytes Data to Flash
---Write 1441792 / 1469959 Bytes Data to Flash
---Write 1469952 / 1469959 Bytes Data to Flash
---Write 1469959 / 1469959 Bytes Data to Flash
Download OK, and Checking the flash data...
Congratulation, download success!
---Click [Upgrade Dog Download]
After the download is complete, the prompt message is as shown in the figure
当前芯片型号选择为:【B1X系列】 - - - Product number

M6.upgradeFlash+BurnKEY+BurnMAC DataSteps

Step6:willKeyAnd download the pattern to the burner

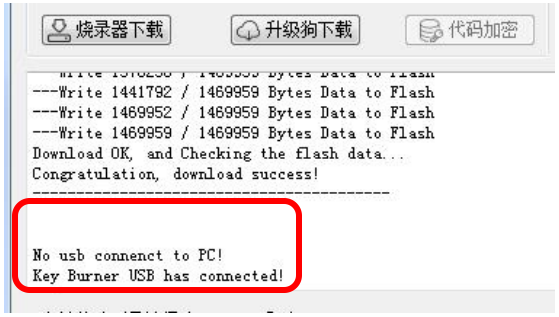


1.Burner appearance

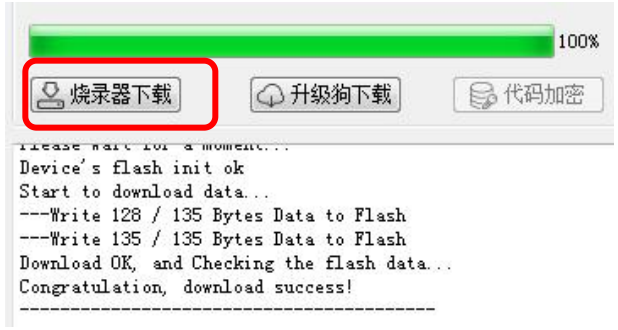




2.Connect to the computer display-PC-



3.After the burner is connectedMV Assistant Show connection



4.Click [Burner Download], a message will be prompted after the download is complete



5.Burner displayEND

M6.upgradeFlash+BurnKEY+BurnMAC DataSteps

Step6:willKeyAnd download the pattern to the burner



6.After the writer is powered on again, it will displayM6 Mode word

M6.upgradeFlash+BurnKEY+BurnMAC DataSteps

Step7: Start upgrading and burning operations



BMouthedUSBConnect

(1) Unplug theBMouthedUSBConnection line,

(2)willFlash_Burn_LiteInsert theUSBinterface

(3) Power on the writer again

(4) After waiting a few seconds, the digital tube displaysnn-6Indicates that the burner has enteredM6Mode, ready to upgrade and burn

(5) Place the chip, after closing the cover, press theStartPress the button to start upgrading and burning immediately,

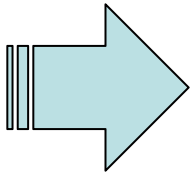
After successful burning, the digital tube will displayP-00

As shown below:


量产应用：
Step1：无需和电脑连接，直接插入烧录器的USB接口
Step2：给烧录器通电，等初始化OK后，按下Start按键即可自动烧录和升级
MVSILICON
START
Flash_Burner_Lite
开始烧录按键
烧录器电源入口

M0.Burn onlyKEY

Instructions



M0.Burn onlyKEY

【Steps】

Step1: Prepare the writer and chip, connect the writer to the computer, and display-PC-Means connectionOK

Step2：Open the host computer toolMVAssistant, Select chip type, mode selectionM0,

Enter the password, click [Burner Download], set the mode andKeyDownload to the burner, the download is successful, the digital tube displaysEndAnd then unplugUSBLine, digital tube displaynn-0

Step3: Then, click [Upgrade Dog Download], andKeyDownload toFlashUpgrading the dog tool

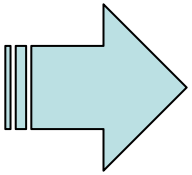
Step4:willFlashInsert the upgrade dog tool into the left side of the burnerUSBInterface, press theStart

Press the key to start burningKey, The operation is successfulP-00

Note: For detailed operation process, please refer to the previous modeM6instruction of

M1.Burn onlyMAC Data

Instructions



M1.Burn onlyMAC Data

【Steps】

Step1: Prepare the writer and chip, connect the writer to the computer, and display-PC-Means connection OK Step2：Open the host computer toolMVAssistant, Select chip type, mode selectionM1,enter BTandBLEName, selectMACThe data input method corresponds toMACEnter data in the data box or load the file, click [Burner Download] to download the data to the writer, if the download is successful, the digital tube displaysEndAnd then unplugUSBLine, digital tube displaynn-1 Step3: Press theStartPress the key to start burningKey, The operation is successfulP-00

instruction:(1) For detailed operation process, please refer to the previous modeM6instruction of

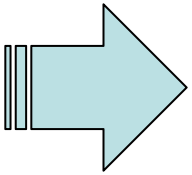
(2) Burn separatelyMAC DataWithoutFlashCoordination of upgrade tools (3) If you chooseInput data, Then every chip is burned the same6byte,

If you chooseconfig file, The file is loaded, each data to be burned is different, and each chip burns different data, and a file in a specific format is required

(4) Due to burningKeyAfter the chipSWIs banned, so try to burn firstMAC Data

M2.Upgrade onlyFlash

Instructions



M2.Upgrade onlyFlash

【Steps】

Step1: Prepare the writer and chip, connect the writer to the computer, and display-PC-Means connectionOK

Step2：Open the host computer toolMVAssistant, Select chip type, mode selectionM2,

Click [Download from Burner] to download the model to the burner, the download is successful, and the digital tube displaysEndAnd then unplugUSBLine, digital tube displaynn-2

Step3: Next, loadCodeFile, click [Upgrade Dog Download] to download the file data to FlashUpgrading the dog tool

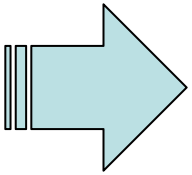
Step4:willFlashInsert the upgrade dog tool into the left side of the burnerUSBInterface, press the Start Press the button to start the upgradeFlash, The operation is successfulP-00

instruction:(1) For detailed operation process, please refer to the previous modeM6instruction of

(2) If you want to upgrade the encryptedCode,existStep3Insert dongle pairCodeencryption

M3.BurnKEY+BurnMAC Data

Instructions



M3.BurnKEY+BurnMAC Data

【Steps】

Step1: Prepare the writer and chip, connect the writer to the computer, and display-PC-Means connectionOK

Step2：Open the host computer toolMVAssistant, Select chip type, mode selectionM3, enterKeyAnd Bluetooth name and address data, click [Burner download], the mode and Key+MAC DataDownload to the burner, the download is successful, the digital tube displaysEnd, Then unplugUSBLine, digital tube displaynn-3

Step3: Then, click [Upgrade Dog Download], andKeyData download toFlashUpgrading the dog tool

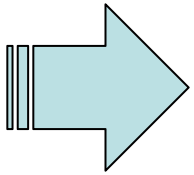
Step4:willFlashInsert the upgrade dog tool into the left side of the burnerUSBInterface, press theStart Press the key to start burningKeyandMAC Data, The operation is successfulP-00

instruction:(1) For detailed operation process, please refer to the previous modeM6instruction of

(2)NoticeMAC dataFor the loading method, please refer to the previous description (3) Only when burning separatelyMAC dataDo not insertFlashupgrade tools

M4.upgradeFlash+BurnKEY

Instructions



M4.upgradeFlash+BurnKEY

【Steps】

Step1: Prepare the writer and chip, connect the writer to the computer, and display-PC-Means connectionOK

Step2：Open the host computer toolMVAssistant, Select chip type, mode selectionM4,enter KEYValue, click [Burner Download], andKEYThe value plus is downloaded to the burner, the download is successful, and the digital tube displaysEndAnd then unplug USBLine, digital tube displaynn-4

Step3：Next toPCInsert the dongle, then load the code, click to upgrade to download, prompting that the encryption is complete After finishing, click [Upgrade Dog Download], andKEYandCodeDownload toFlashUpgrade dog tool middle

Step4:willFlashInsert the upgrade tool into the left side of the burnerUSBInterface, press theStart Press the button to start the upgradeFlashAnd burnKEYValue, the operation is successfulP-00

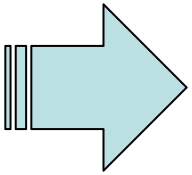
instruction:(1) For detailed operation process, please refer to the previous modeM6instruction of

(2)NoticeMAC DataFor the loading method, please refer to the previous description (3)

Only when burning separatelyMAC DataDo not insertFlashupgrade tools

M5.upgradeFlash +BurnMAC Data

Instructions



M5.upgradeFlash+BurnMAC Data

【Steps】

Step1: Prepare the writer and chip, connect the writer to the computer, and display-PC-Means connectionOK

Step2：Open the host computer toolMVAssistant, Select chip type, mode selectionM5, Enter the Bluetooth name and address, click [Burner Download], and set the mode andMAC DataDownload to the burner, the download is successful, the digital tube displaysEndAnd then unplugUSBLine, digital tube displaynn-5

Step3: Next, loadCodeFile, click [Upgrade Dog Download] to download the data toFlash

Upgrading the dog tool

Step4:willFlashInsert the upgrade tool into the left side of the burnerUSBInterface, press theStart

Press the button to start the upgradeFlashAnd burnMAC Data, The operation is successfulP-00

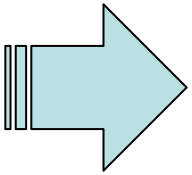
instruction:(1) For detailed operation process, please refer to the previous modeM6instruction of

(2)NoticeMAC DataFor the loading method, please refer to the previous description (3)

Only when burning separatelyMAC DataDo not insertFlashupgrade tools

M7.【KEYcheck】

Instructions



M7.Check onlyKEY

【Steps】

Step1: Prepare the writer and chip, connect the writer to the computer, and display-PC-Means connectionOK

Step2：Open the host computer toolMVAssistant, Select chip type, mode selectionM7,

Enter the password, click [Burner Download], set the mode andKeyDownload to the burner, the download is successful, the digital tube displaysEndAnd then unplugUSBLine, digital tube displaynn-7

Step3: Then, click [Upgrade Dog Download], andKeyDownload toFlashUpgrade tool

Step4:willFlashInsert the upgrade tool into the left side of the burnerUSBInterface, press theStart

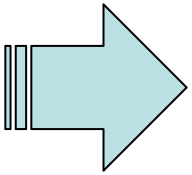
Press the key to startcheckKey, The operation is successfulP-00

instruction:(1) For detailed operation process, please refer to the previous modeM4instruction of

(2) The modeM7And patternM0, The operation is the same, but the modeM0Need to KeyBurned into the chip, andM7Mode only for chipKeyDo the verification, not the burning action.

M8.【upgradeFlash+Equipment name】

Instructions



M8.upgradeFlash+Equipment name

【Steps】

Step1: Prepare the writer and chip, connect the writer to the computer, and display-PC-Means connectionOK

Step2：Open the host computer toolMVAssistant, Select chip type, mode selectionM8, Enter the Bluetooth name and address, click [Burner Download], download the mode and device name to the writer, the download is successful, the digital tube displaysEndAnd then unplug USBLine, digital tube displaynn-8

Step3: Next, loadCodeFile, click [Upgrade Dog Download] to download the data toFlashUpgrading the dog tool

Step4:willFlashInsert the upgrade tool into the left side of the burnerUSBInterface, press theStart

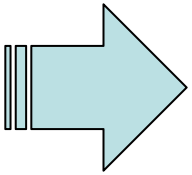
Press the button to start the upgradeFlashAnd the name of the burning device, the operation is successfulP-00

instruction:(1) For detailed operation process, please refer to the previous modeM6instruction of

(2)NoticeMAC dataFor the loading method, please refer to the previous description (3) Only when burning separatelyMAC dataDo not insertFlashupgrade tools

M9.【upgradeFlash+BurnKEY+Equipment name】

Instructions



M9.upgradeFlash+BurnKEY+Equipment name

【Steps】

Step1: Prepare the writer and chip, connect the writer to the computer, and display-PC-Means connectionOK

Step2：Open the host computer toolMVAssistant, Select chip type, mode selectionM9,enterKEYValue, click [Burner Download], andKEYThe value and device name are downloaded to the burner, and the digital tube will display when the download is successful.

EndAnd then unplugUSBLine, digital tube displaynn-9

Step3：Next toPCInsert the dongle, then load the code, click upgrade to download, and after the prompt is completed, click [Upgrade Dog Download], andKEYandCodeDownload toFlashUpgrading the dog tool

Step4:willFlashInsert the upgrade tool into the left side of the burnerUSBInterface, press theStartPress the button to start the upgradeFlashAnd burnKEYValue and device name, the operation is successful displayP-00

instruction:(1) For detailed operation process, please refer to the previous modeM6instruction of

(2)NoticeMAC DataFor the loading method, please refer to the previous description (3)

Only when burning separatelyMAC DataDo not insertFlashupgrade tools

3. List of error codes and problem handling

Schedule I List of common error codes

hint:P-XX Indicates operation [normal]

E-XX Indicates operation [abnormal]

Display code	definition		Solution
P-00	M7 mode:	It is detected that the Key in the chip is empty	Normal operation
	M0~M6 mode: Indicates that the programming or upgrading operation is normal in this mode		Normal operation
P-01	M0 mode:	It is detected that the burned key in the chip is the same as the key to be burned	Normal operation
	M2 mode:	The upgrade is successful, the key is detected	Normal operation
	M3 mode:	Key and MAC Data have been burned	Normal operation
	M4 mode:	The upgrade is successful, it is detected that the Key has been burned	Normal operation
	M6 mode:	The upgrade is successful. It is detected that the Key and MAC Data have been burned. It is	Normal operation
	M7 mode:	detected that the burned key in the chip is the same as the key to be burned.	Normal operation

Display code	definition	Solution
E-20	The writer failed to download data	Download the data again
E-25	System Flash unlock failed	System error, restart the writer
E-30	The chip is not detected or the chip position is not placed correctly	Check whether the chip position is placed
E-31	SW initialization failed when burning efuse	Try to press repeatedly
E-32	PLL lock failed when burning efuse	Try to press repeatedly
E-33	No FT logo detected	Contact MVSilicon
E-42	Inconsistent burned keys are detected when burning key mode	May be mixed, please make sure it is the same batch of chips
E-44	Key or MAC Data programming error	Burning failed, please replace the chip and burn again
E-50	Failed to find the next data when using file to input MAC Data.	Check whether the data file meets the requirements
E-51	Inconsistent MAC Data has been burned.	May be mixed, please make sure it is the same batch of chips
E-60	Mode error	Re-download data and patterns
E-61	Flash upgrade failed	The upgrade failed, you can try to press the key repeatedly
E-62	It is detected that the chip has been burned when burning the MAC Data related mode. Key	Cannot burn MAC Data after burning the key
E-63	Flash upgrade timeout	Check Code size
E-64	Uart communication failed, flash burner was not plugged	Check if the flash burner is plugged in
E-65	in and received wrong result data	Try to press repeatedly

Schedule two **List of error codes in various modes of operation**

Operation mode	Display code	Means and solutions
M0: Burn only KEY	P-00	1. Representative meaning: normal, the chip burns the key for the first time 1. Solution: None
	P-01	2. Representative meaning: normal, the chip burns the key repeatedly 2. Solution: None
	E-42	3. Representative meaning: an error is reported, the chip has burned the key, and it is inconsistent with the key to be burned this time 3. Solution: Please download the key again, and use an empty film to burn the key
	E-60	4. Representative meaning: error, there is code in Flash Burner Lite 4. Solution: In this mode, you can only burn the Key, so you need to download the Key data again, don't load the Code content
	E-64	5. On behalf of the meaning: an error is reported, the flash burner is not inserted or the uart communication fails 5. Solution: Please re-insert the Flash Burner Lite tool, check the power supply and whether the connection is normal
M1: Burn only MAC Data	P-00	1. Representative meaning: normal, blank chip burn MAC Data is normal 1. Solution: None
	P-01	2. Representative meaning: normal, same MAC Data after re-burning 2. Solution: None
	E-60	3. Representative meaning: an error was reported, and the key was burned 3. Solution: In this mode, only MAC Data can be burned, so MAC Data needs to be re-downloaded instead of Key content
	E-51	4. Representative meaning: report an error, burn different MAC Data 4. Solution: The chip has burned MAC Data before, and the MAC Data to be burned this time Inconsistent, the user needs to update the chip that has not burned MAC Data to burn

Operation mode	Display code	Means and solutions
M2: Upgrade Flash only	P-00	1. Representative meaning: normal, upgrade Flash with empty film 1. Solution: None
	P-01	2. Representative meaning: normal, the chip with the key has been upgraded 2. Solution: None
	E-64	3. Represents meaning: an error is reported, the flash burner is not inserted or the uart communication fails 3. Solution: Please re-insert the Flash Burner Lite tool, check the power supply and whether the connection is normal
	E-60	4. Representative meaning: error, no code data in Flash Burner 4. Solution: In this mode, only Flash can be upgraded, so Code data needs to be downloaded again
	E-61	5. On behalf of the meaning: error, Flash upgrade failed 5. Solution: try the upgrade operation again, or replace the chip to test
M3: Burn Key + Burn MAC Data	P-00	1. Representation meaning: normal, burn Key+MAC Data with empty film 1. Solution: None
	P-01	2. Representative meaning: normal, the chip that has been burned correctly is reburned 2. Solution: None
	E-42	3. Representative meaning: an error is reported, and chips with inconsistent keys are re-burned 3. Solution: The chip has burned the key before, and it is inconsistent with the key burned this time
	P-00	4. Representative meaning: normal, the chip with the correct MAC Data has been burned again, no Key 4. Solution: The chip has burned MAC Data before, and the Key and MAC Data are burned correctly this time
	P-01	5. Representative meaning: normal, re-burning of chips that have already burned the same Key 5. Solution: The chip has burned the Key before, and the burned Key and MAC Data are correct this time
	E-44	6. Representative meaning: error, failed to burn Key or MAC Data 6. Solution: try the upgrade operation again, or replace the chip to test
M4: Upgrade Flash + burning KEY	P-00	1. Representative meaning: normal, blank film burning + upgrade 1. Solution: None
	P-01	2. Representative meaning: normal, the chip with the same key has been burned 2. Solution: None
	E-42	3. Representative meaning: report an error, burn chips with inconsistent keys 3. Solution: replace the chip and reprogram, or check whether the Key value is correct
	E-60	4. Representative meaning: error, no code data in Flash Burner 4. Solution: Please re-use the host computer tool to download Code data to Flash_Burner_Lite
	E-44	5. On behalf of the meaning: an error is reported, the key burning failed 5. Solution: replace the chip and reprogram
	E-61	6. On behalf of the meaning: error, Flash upgrade failed 6. Solution: try the upgrade operation again, or replace the chip to test
	E-64	7. On behalf of the meaning: error, flash burner is not inserted or uart communication failure 7. Solution: Please reinsert the Flash Burner Lite tool, check the power supply and whether the connection is normal

Operation mode	Display code	Means and solutions
M5: Upgrade Flash + Burn MAC Data	P-00	1. Representative meaning: normal, blank film burned 1. Solution: None
	E-62	2. Representative meaning: report an error, burn the chip with the key burned 3. Solution: The chip has already burned the Key, and then burning the MAC Data again will definitely fail, and this kind of operation is not allowed. Either burn the MAC Data first, or burn the Key and MAC Data at the same time
	P-01	3. Representative meaning: normal, re-burning of chips with consistent MAC Data 3. Solution: Please check the MAC Data content, or replace the chip and burn again
	E-51	4. Representative meaning: report an error, and re-burn the chip with inconsistent MAC Data. 4. Solution: Please check the MAC Data content, or replace the chip and burn again
	E-60	5. On behalf of the meaning: an error is reported, no Code is detected in Flash Burner 5. Solution: Please re-use the host computer tool to download Code data to Flash_Burner_Lite
	E-64	6. On behalf of the meaning: an error is reported, the flash burner is not inserted or the uart communication fails 6. Solution: Please re-insert the Flash Burner Lite tool, check the power supply and whether the connection is normal
	E-44	7. Representative meaning: an error is reported, and the chip has failed to burn MAC Data 7. Solution: try the upgrade operation again, or replace the chip to test
	E-61	8. Representative meaning: error, chip upgrade FlashCode failed 8. Solution: try the upgrade operation again, or replace the chip to test

Operation mode	Display code	Means and solutions
M6: Upgrade Flash + Burn Key + Burn MAC Data	P-00	1. Representative meaning: normal, blank film burned 1. Solution: None
	P-01	2. Representative meaning: normal, burned phase Re-burning of the same key chip 2. Solution: None
	E-42	3. Representative meaning: report an error, have not burned Re-burning of chips with consistent Key 3. Solution: The chip has been burned before Key, and it is inconsistent with the key burned this time
	P-00	4. Representative meaning: normal, only burned Re-burning of the same MAC Data chip 4. Solution: The chip has been burned before MAC Data, burn the key again this time, MAC Data and upgrade are normal,
	E-60	5. On behalf of the meaning: error, Flash B no Code is detected in the urner 5. Solution: Please re-use the upper position Downloading Code data to Flash_Burner_Lite by the machine tool,
	E-64	6. Representative meaning: an error is reported, f is not inserted lash burner or uart communication failed 6. Solution: Please re-insert Fla sh Burner Lite tool, check the power supply and whether the connection is normal.
	E-44	7. Representative meaning: error, Key or MA C Data burning failed 7. Solution: Replace the chip
	E-61	8. On behalf of the meaning: error, Flash upgrade failed 8. Solution: try the upgrade operation again, or replace the chip to test
M7: KEY check	P-00	1. Representative meaning: normal, the chip is empty, and the key has not been burned 1. Solution: None
	P-01	2. Representation meaning: normal, the chip has burned the key, and is consistent with the verified key 2. Solution: None
	E-42	3. Representative meaning: an error is reported, the chip has already burned the Key, and is inconsistent with the verified Key 3. Solution: It can be determined that the Key of the chip under test is inconsistent with the Key of verification
	E-64	4. Represents meaning: an error is reported, the flash burner is not inserted or the uart communication fails 4. Solution: Please re-insert the Flash Burner Lite tool, check the power supply and whether the connection is normal

Operation mode	Display code	Means and solutions
M8: Upgrade Flash + Equipment name	P-00	1. Representative meaning: normal, blank film burned 1. Solution: None
	P-01	2. Representative meaning: normal, burned phase Re-burning of the same key chip 2. Solution: None
	E-42	3. Representative meaning: report an error, have not burned Re-burning of chips with consistent Key 3. Solution: The chip has been burned before Key, and it is inconsistent with the key burned this time
	P-00	4. Representative meaning: normal, only burned Re-burning of the same MAC Data chip 4. Solution: The chip has been burned before MAC Data, burn the key again this time, MAC Data and upgrade are normal,
	E-60	5. On behalf of the meaning: error, Flash B no Code is detected in the urner 5. Solution: Please re-use the upper position Downloading Code data to Flash_Burner_Lite by the machine tool,
	E-64	6. Representative meaning: an error is reported, f is not inserted lash burner or uart communication failed 6. Solution: Please re-insert Fla sh Burner Lite tool, check the power supply and whether the connection is normal.
	E-44	7. Representative meaning: error, Key or MA C Data burning failed 7. Solution: Replace the chip
	E-62	8. Represents meaning: an error is reported, the chip is already The key is burned, and the device name cannot be burned 8. Solution: After burning the Key, the flash cannot be burned by the burner
M9: Upgrade Flash + Burn Key + Equipment name	P-00	1. Representative meaning: normal, the chip is empty, and the key has not been burned 1. Solution: None
	P-01	2. Representation meaning: normal, the chip has burned the key, and is consistent with the verified key 2. Solution: None
	E-42	3. Representation meaning: an error is reported, the chip has burned the key, and the key values of the burner, the upgrade dog, and the tested chip are inconsistent or the chip has not burned the key, but the upgrade dog and the burner download the key are inconsistent 3. Solution: Please download the key again, use the blank film to burn the key, check whether the upgrade dog and the burner's key value are the same, you can observe the burner print information to distinguish, judge which party downloaded the key incorrectly
	E-64	4. Represents meaning: an error is reported, the flash burner is not inserted or the uart communication fails 4. Solution: Please re-insert the Flash Burner Lite tool, check the power supply and whether the connection is normal

Four, update records

Tool update record

serial number	PC version	Update content description
1	MVAssistant V1.7.5.3	It can be used for BP1xx series chips to burn Key, upgrade Flash, burn Bluetooth address, and device name (2019.07.23)
2	MVAssistant V1.7.7	Increase M8, M9 mode (2019.08.23)